

⑦

Convert $z = -2 + 2\sqrt{3}j$

to polar form and to exponential form. Give the modulus and principal argument.

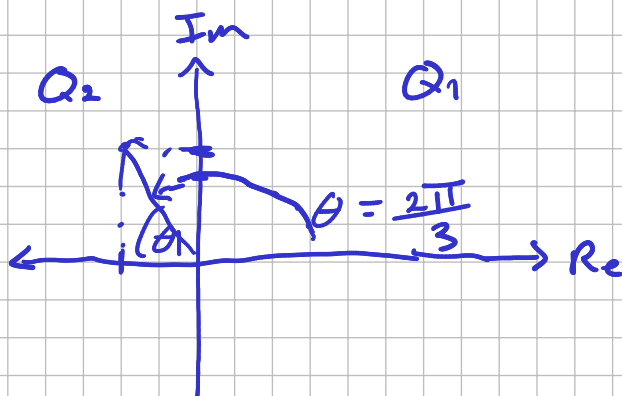
Define terms: (oxfordreference.com)

Polar form: $z = r(\cos \theta + j \sin \theta)$ Exponential: $z = re^{j\theta}$ Modulus: $|z| = r$ Argument: θ

1) Get the modulus:

$$\begin{aligned}
 r &= \sqrt{(-2)^2 + (2\sqrt{3})^2} \\
 &= \sqrt{4 + (2\sqrt{3})(2\sqrt{3})} \\
 &= \sqrt{4 + 4 \cdot 3} = \sqrt{16} = \underline{4} \quad \text{modulus}
 \end{aligned}$$

2)



$$\sin \theta = \frac{\text{Im}}{|z|} = \frac{\text{Im}}{r}$$

$$\sin \theta = \frac{2\sqrt{3}}{4}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \sin^{-1} \frac{\sqrt{3}}{2}$$

$$\theta_{\text{ref}} = \frac{\pi}{3}$$

⚠ Careful! we are in Q2!

$$\pi - \frac{\pi}{3} = \frac{3\pi}{3} - \frac{\pi}{3} = \underline{\underline{\frac{2\pi}{3}}} \quad \text{Argument}$$

Polar form: $z = 4 \left[\cos\left(\frac{2\pi}{3}\right) + j \sin\left(\frac{2\pi}{3}\right) \right]$

Exponential Form: $4e^{j2\pi/3}$